Abdomen Anatomy & Viscera Overview

**(introduction of abdomen
Quadrants, regions and the introduction of oesophagus,
stomach, small and large intestine, pancreas, liver and spleen)**

1. The abdominal cavity is separated from the thoracic cavity superiorly by a major, dome-shaped musculotendinous partition. This partition is the:
a. Diaphragm
b. Pelvic diaphragm
c. Transversus abdominis
d. Quadratus lumborum
e. Linea alba
2. To simplify clinical descriptions, the abdomen is divided into four quadrants by two imaginary planes intersecting at the umbilicus. These planes are the:
a. Median (sagittal) and transumbilical planes
b. Transpyloric and transtubercular planes
c. Subcostal and interspinal planes
d. Midclavicular and horizontal planes
e. Coronal and sagittal planes
3. For a more detailed topographical mapping, the abdomen is divided into nine regions by two vertical and two horizontal planes. The superior horizontal plane is the:
a. Subcostal plane (or transpyloric plane)
b. Transtubercular plane
c. Interspinal plane
d. Transumbilical plane
e. Intercristal plane
4. The inferior horizontal plane used to map the nine abdominal regions connects the tubercles of the iliac crests, formally designated as the:
a. Subcostal plane
b. Transpyloric plane
c. Midclavicular plane
d. Transtubercular plane
e. Interspinal plane
5. The two vertical planes used to divide the abdomen into nine distinct zones pass downward from the midpoint of each clavicle, known as the:
a. Parasagittal planes
b. Midaxillary planes
c. Transtransverse lines
d. Right and left midclavicular planes
e. Anterior median lines
6. The central or middle region among the nine abdominal zones, located directly surrounding the umbilicus, is the:
a. Epigastric region
b. Umbilical region
c. Hypogastric region

d. Right lumbar region
e. Left iliac fossa

7. Situated immediately superior to the umbilical region, bounded laterally by the right and left hypochondriac regions, is the:

a. Epigastric region
b. Hypogastric region
c. Pubic region
d. Left lumbar zone
e. Right iliac region

8. Located inferior to the umbilical region and bounded between the right and left iliac (inguinal) regions is the pubic region, also termed the:

a. Epigastrium
b. Left flank
c. Hypogastric region
d. Right flank
e. Right hypochondrium

9. Let's introduce the viscera. The abdominal part of the esophagus is a short muscular segment that enters the abdomen by piercing the muscular diaphragm at which vertebral level?

a. 8th thoracic vertebra (T8)
b. 10th thoracic vertebra (T10)
c. 12th thoracic vertebra (T12)
d. 1st lumbar vertebra (L1)
e. 5th thoracic vertebra (T5)

10. After entering the abdomen, the esophagus runs a short course of about 1 to 2 cm to terminate by emptying into the stomach at the:

a. Pyloric sphincter
b. Fundus gateway
c. Angular notch
d. Cardiac orifice
e. Duodenojejunal flexure

11. The stomach is a dilated, J-shaped segment of the digestive tract. It features two openings, two surfaces, and two distinct borders known as the:

a. Superior and inferior notch lines
b. Lesser and greater curvatures
c. Anterior and posterior poles
d. Medial and lateral ridges
e. Longitudinal and horizontal lines

12. The stomach is divided into zones. The dome-shaped upward projection situated to the left of the cardiac orifice and filled with gas on a standing radiograph is the:

a. Body
b. Pyloric antrum
c. Pyloric canal
d. Fundus
e. Lesser omentum bed

13. The main, massive central portion of the stomach located between the fundus and the pyloric antrum is the:

a. Pylorus

b. Cardiac antrum
c. Body (corpus)
d. Angular notch line
e. Rugae matrix

14. The terminal, narrow muscular exit of the stomach that regulates the passage of chyme into the duodenum is guarded by the:

a. Cardiac sphincter
b. Ileocecal valve
c. Internal anal sphincter
d. Pyloric sphincter
e. Valve of Luschka

15. When the stomach is empty, its internal mucosal lining is thrown into numerous temporary, longitudinal ridges and folds known as:

a. Villi
b. Plicae circulares
c. Rugae
d. Haustra
e. Taeniae

16. The small intestine is the primary site for nutrient absorption. Macroscopically, it is divided sequentially into three distinct segments, which are the:

a. Esophagus, stomach, and colon
b. Duodenum, jejunum, and ileum
c. Cecum, colon, and rectum
d. Fundus, body, and pylorus
e. Ascending, transverse, and descending segments

17. The first, shortest, and widest segment of the small intestine is a C-shaped loop that wraps around the head of the pancreas, known as the:

a. Jejunum
b. Duodenum
c. Ileum
d. Cecum
e. Cardia

18. The duodenum is unique because unlike the rest of the small intestine, most of its course is fixed to the posterior abdominal wall, making it:

a. Intraperitoneal
b. Retroperitoneal
c. Subperitoneal
d. Mobile with a mesentery
e. Suspended inside the lesser omentum

19. The second, mobile segment of the small intestine, constituting roughly two-fifths of its total length and occupying mainly the upper left quadrant of the abdomen, is the:

a. Ileum
b. Jejunum
c. Duodenum
d. Appendix
e. Colon

20. The third, highly coiled segment of the small intestine, constituting the distal three-fifths of its length and ending at the ileocecal valve, is the:
a. Duodenum
b. Jejunum
c. Cecum
d. Ileum
e. Sigmoid loop
21. The jejunum and ileum are suspended from the posterior abdominal wall by a broad, fan-shaped double layer of peritoneum that carries their neurovascular bundles, termed:
a. The greater omentum
b. The lesser omentum
c. The mesentery
d. The falciform ligament
e. The epiploic appendages
22. The large intestine extends from the ileocecal junction to the anus. It is easily distinguished macroscopically from the small intestine by the presence of sacculations or pouches called:
a. Rugae
b. Plicae semilunares
c. Villi
d. Haustra
e. Peyer's patches
23. Running along the length of the cecum and colon are three narrow, distinct longitudinal ribbons of smooth muscle, known as the:
a. Appendices epiploicae
b. Taeniae coli
c. Plicae circulares
d. Valvulae conniventes
e. Rugae cords
24. Projecting from the serosal surface of the colon are small, isolated, fat-filled pouches of peritoneum unique to the large intestine, classified as:
a. Haustra
b. Villi pads
c. Appendices epiploicae (omental appendages)
d. Lymphoid nodules
e. Lacteal buds
25. The large intestine begins in the right iliac fossa as a blind-ending, wide pouch situated inferior to the ileocecal valve, known as the:
a. Ascending colon
b. Rectum
c. Cecum
d. Appendix
e. Sigmoid colon
26. Arising from the posteromedial wall of the cecum, roughly 2 cm inferior to the ileocecal valve, is a narrow, worm-like diverticulum rich in lymphoid tissue, named the:
a. Meckel's diverticulum

b. Vermiform appendix
c. Lingula
d. Gallbladder
e. Spleen hilum

27. The colon is divided into structural zones. The segment that ascends vertically from the cecum along the right side of the abdominal cavity to reach the liver is the:
a. Transverse colon
b. Descending colon
c. Ascending colon
d. Sigmoid colon
e. Rectum

28. At the lower surface of the liver, the ascending colon turns sharply to the left to continue across the abdomen. This sharp right turn is the:
a. Splenic flexure
b. Left colic flexure
c. Hepatic flexure (right colic flexure)
d. Duodenojejunal flexure
e. Pyloric notch

29. The longest and most mobile segment of the large intestine arches completely across the abdomen from the right hypochondrium to the left hypochondrium. This is the:
a. Ascending colon
b. Transverse colon
c. Descending colon
d. Sigmoid colon
e. Rectum

30. Below the spleen, the transverse colon turns sharply downward to become the descending colon. This left-sided structural turn is the:
a. Hepatic flexure
b. Right colic flexure
c. Splenic flexure (left colic flexure)
d. Ileocecal valve
e. Cardiac incisura

31. The descending colon runs down the left side of the abdomen and passes into the pelvis as a mobile, S-shaped loop called the:
a. Rectum
b. Cecum
c. Anal canal
d. Sigmoid colon
e. Ascending colon

32. The rectum is a fixed, dilated segment of the large intestine that sits inside the pelvic cavity, anterior to the sacrum. It completely lacks which feature characteristic of the colon?
a. A muscular wall
b. An internal mucosal lining
c. Taeniae coli, haustra, and appendices epiploicae
d. Blood vessels and nerves
e. Peritoneal coverage anywhere

33. The liver is the largest gland in the human body. Macroscopically, it occupies almost the entirety of which abdominal zone?
a. Left iliac fossa
b. Hypogastric region
c. Right hypochondrium (along with parts of the epigastrium and left hypochondrium)
d. Left lumbar flank
e. Pubic area
34. The diaphragmatic surface of the liver is smooth and convex, matching the curvature of the diaphragm. It is divided into large right and left structural lobes by a crescentic dural-like peritoneal fold called the:
a. Lesser omentum
b. Falciform ligament
c. Ligamentum teres
d. Ligamentum venosum
e. Coronary ligament
35. The visceral (inferior) surface of the liver features several high-yield structural markings. Located on this surface are two small accessory lobes named the:
a. Temporal and frontal lobes
b. Eparterial and hyparterial lobes
c. Quadrate and caudate lobes
d. Pyramidal and lingular lobes
e. Medial and lateral lobes
36. The porta hepatis is a deep, transverse fissure on the visceral surface of the liver. It acts as the gateway where which primary neurovascular structures enter or leave the liver parenchyma?
a. Hepatic veins and inferior vena cava
b. Portal vein, hepatic artery proper, and hepatic bile ducts
c. Splenic artery and vein
d. Internal thoracic artery and phrenic nerve
e. Celiac trunk and vagus nerve trunks
37. Tucked securely inside a shallow fossa on the inferior surface of the right hepatic lobe is a pear-shaped sac that stores and concentrates bile, known as the:
a. Pancreas
b. Spleen
c. Appendix
d. Gallbladder
e. Duodenum
38. The liver receives a unique, dual blood supply. Approximately 75% to 80% of its total blood volume is delivered via a major vein that collects nutrient-rich blood from the GIT, known as the:
a. Hepatic vein
b. Inferior vena cava
c. Portal vein (vena portae)
d. Splenic vein
e. Superior mesenteric vein
39. The remaining 20% to 25% of highly oxygenated arterial blood is supplied to the liver tissue via the:
a. Celiac trunk directly

b. Hepatic artery proper
c. Superior mesenteric artery
d. Inferior thyroid artery
e. Renal artery trunk

40. Venous blood leaving the liver sinusoids is collected by several large intrahepatic channels that empty directly into the adjacent inferior vena cava. These channels are the:

a. Portal veins
b. Hepatic veins
c. Splenic veins
d. Superior mesenteric veins
e. Pericardial veins

41. The pancreas is an elongated, soft accessory digestive gland featuring both exocrine and endocrine functions. Structurally, it is divided from right to left into a:

a. Base, body, and apex
b. Head, neck, body, and tail
c. Anterior and posterior arch
d. Superior and inferior pole
e. Main and accessory lobe

42. The head of the pancreas is the widest part of the gland. It is nestled snugly within the concavity of which surrounding structural framework?

a. The lesser curvature of the stomach
b. The C-shaped loop of the duodenum
c. The hilum of the spleen
d. The visceral surface of the left lobe of the liver
e. The pelvic inlet ring

43. Extending from the lower left portion of the pancreatic head is a small, hook-like projection that passes posterior to the superior mesenteric vessels, termed the:

a. Pyramidal process
b. Uncinate process
c. Lingula
d. Accessory tail
e. Hamulus

44. The narrow, left-most extremity of the pancreas is mobile and tracks laterally to terminate within the hilum of which adjacent organ?

a. Left kidney
b. Gallbladder
c. Spleen
d. Left lobe of liver
e. Descending colon

45. The exocrine secretions of the pancreas (digestive enzymes and bicarbonate) are collected by a long, horizontal channel traversing the length of the gland, known as the:

a. Common bile duct
b. Main pancreatic duct (of Wirsung)
c. Cystic duct
d. Stensen's duct
e. Wharton's duct

46. Near its termination, the main pancreatic duct joins the common bile duct to form a short, dilated dilation that opens into the second part of the duodenum, formally called the:
a. Ampulla of Vater (hepatopancreatic ampulla)
b. Fossa of Rosenmüller
c. Valve of Kerckring
d. Crypt of Lieberkühn
e. Pyloric gate
47. The opening of the hepatopancreatic ampulla into the duodenal lumen forms a small, visible mucosal elevation guarded by a sphincter muscle (sphincter of Oddi). This structural elevation is the:
a. Minor duodenal papilla
b. Major duodenal papilla
c. Plica semilunaris
d. Ileocecal valve lip
e. Carina
48. The endocrine portion of the pancreas consists of millions of microscopic clusters of cells scattered among the exocrine acini, responsible for secreting insulin and glucagon. These clusters are the:
a. Peyer's patches
b. Hassall's corpuscles
c. Islets of Langerhans
d. Corpora arenacea
e. Salivary acini
49. The spleen is a soft, highly vascular, purple-colored lymphoid organ. Macroscopically, it is located in the upper left quadrant of the abdomen, completely tucked inside the:
a. Right hypochondrium
b. Left hypochondrium (level with ribs 9 through 11)
c. Hypogastric region
d. Right iliac fossa
e. Central umbilical zone
50. The spleen is bounded anteriorly by the stomach, posteriorly by the left diaphragm, and inferiorly by the left colic flexure. Its smooth, diaphragmatic surface is separated from its deeply grooved:
a. Visceral surface (featuring the hilum for the splenic artery and vein)
b. Superior pole apex
c. Anterior median line
d. Lesser curvature margin
e. Pyloric margin
51. Because the spleen is a soft, fragile organ packed with blood, a fracture of the overlying 9th, 10th, or 11th ribs on the left side can easily lacerate its capsule, resulting in profuse, life-threatening:
a. Salivary hypersecretion
b. Intracranial pressure shifts
c. Intraperitoneal hemorrhage (severe internal bleeding)
d. Air embolism into the lungs
e. Bile leaks into the stomach

52. The celiac trunk is a massive, short branch arising from the ventral aspect of the abdominal aorta. It is the primary arterial highway supplying all derivatives of which embryonic segment?
a. Midgut
b. Hindgut
c. Foregut
d. Respiratory bud
e. Neural crest
53. The celiac trunk immediately splits into three major high-yield branches. These are the:
a. Superior thyroid, lingual, and facial arteries
b. Left gastric, common hepatic, and splenic arteries
c. Superior mesenteric, inferior mesenteric, and renal arteries
d. Internal thoracic, vertebral, and suprascapular arteries
e. Maxillary, mandibular, and superficial temporal arteries
54. The splenic artery is the largest branch of the celiac trunk. It takes a highly tortuous, horizontal path along the superior border of which organ to reach the splenic hilum?
a. Stomach lesser curvature
b. Duodenum arch
c. Pancreas (along its upper body and tail)
d. Left lobe of liver
e. Transverse colon
55. The portal vein is formed behind the neck of the pancreas by the structural union of which two major venous channels?
a. Hepatic vein and inferior vena cava
b. External jugular vein and subclavian vein
c. Splenic vein and superior mesenteric vein
d. Internal iliac vein and renal vein
e. Anterior jugular vein and facial vein
56. A patient presents with a severe blockage of the portal venous system due to liver scarring (cirrhosis). This causes high-pressure backflow called portal hypertension, leading to dilated, varicose veins at the lower end of the esophagus, known as:
a. Hemorrhoids
b. Caput medusae
c. Esophageal varices (which can rupture and cause fatal bleeding)
d. Zenker's diverticula
e. Bronchial mucoceles
57. Portal hypertension can also cause a prominent radials wheel of dilated varicose veins tracking superficially around the patient's umbilicus, a clinical presentation termed:
a. Esophageal varices
b. Caput medusae
c. Hemorrhoids
d. Torticollis
e. Horner's syndrome
58. Varicose dilations of the submucosal venous plexuses at the junction of the rectum and anal canal due to chronic portal backflow or systemic straining are:

a. Esophageal varices
b. Caput medusae
c. Hemorrhoids (piles)
d. Crypts of Lieberkühn
e. Branchial fistulae

59. The dynamic peritoneal fold that suspends the liver from the anterior abdominal wall and the diaphragm features a thick, free lower margin containing a fibrous cord. This cord is the ligamentum teres, which represents an embryonic remnant of the closed:

a. Fetal ductus arteriosus
b. Fetal left umbilical vein
c. Fetal ductus venosus
d. Foramen ovale
e. Vitelline duct stalk

60. The highly protective, extensive double layer of peritoneum that hangs down from the greater curvature of the stomach like an apron, draping over the loops of the jejunum and ileum before folding back to adhere to the transverse colon, is the:

a. Lesser omentum
b. Greater omentum ("policeman of the abdomen")
c. Mesentery proper
d. Falciform fold
e. Gastrosplenic ligament

Answer Key Table (Abdomen & Viscera Overview)

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40	Q 41-48	Q 49-56	Q 57-60
a	b	b	c	c	b	b	b
a	d	b	b	b	b	b	c
b	b	b	c	b	b	c	b
d	c	c	b	c	c	b	b
d	c	c	c	c	b	c	
b	c	b	c	b	a	c	
a	b	c	c	d	b	b	
b	d	b	b	b	c	b	